

Jeffery Liu

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Penultimate-year Mechatronics Engineering student specialising in mechanical design, with hands-on electrical and controls experience. Real design, build and test experience beyond coursework: an R&D internship at Holmes Solutions with a co-authored technical paper, four years running a licensed vehicle dealership, and a turbocharged engine rebuilt in my garage. Takes a problem from first principles through to a part that works.

EDUCATION

University of Canterbury, BE(Hons) Mechatronics Engineering

2024 - present

- Third year, expected graduation 2027. Second-year grade average: A-.
- Offered two competitive UC entrance scholarships (each around 10 awarded nationally); not taken after gaining residency.
- Coursework: solid mechanics, dynamics, fluid mechanics, controls & vibrations, power electronics, and embedded systems.

EXPERIENCE

Mechatronics Engineering Intern, Holmes Solutions

Nov 2025 - Feb 2026

Roadside & Marine sector, R&D consultancy (453.5 hours)

- Investigated premature mooring rope failures end to end: researched the cause and designed an interchangeable sleeve upgrade handling loads above 50 tonnes, delivering test data toward safer mooring guidance industry-wide.
- Mapped apparatus constraints and verified the design with hand calculations and SolidWorks FEA. Flagged a non-conservative capacity check on the revised load case with my own calculations and escalated it for structural review.
- Produced construction-ready drawings and a staged build plan, then sourced contractor quotes and defined scope, sequencing and resourcing with the contractor before preparing weld joints on site. Also supported a second intern's crash-test bogie build and roadside full-scale crash-test setup.
- Design was built and passed full-scale testing in 2026; co-authored the resulting technical paper, to be presented at an international marine conference in August 2026.

Reference: Ray Keane, Senior Engineer, ray.keane@holmessolutions.com

Founder & Licensed Vehicle Dealer, Jeff's Auto Retail

2022 - 2026

- Sourced, diagnosed, repaired and sold used vehicles end to end, and prepared vehicles to WOF standard.
- Bought and sold 35+ cars solo as a sole trader: pricing, listings, customer relations, bookkeeping, tax and compliance.

Performance Consultant & Tuning Support, Custom Culture & Tuning

2025 - present

- Consult directly with clients I source; quote parts, fixes and achievable power levels from hardware limits, then work with an Auckland-based tuner on datalogging and mechanical adjustments to deliver the tune.

Bicycle Technician & Maintenance Tutor, Adventure Cycles

2019 - 2023

- First and longest-running of several bike workshop roles; diagnosed and repaired drivetrain, brake and suspension systems; ran monthly public maintenance workshops, both here and at school through my TravelWise leadership role.

PROJECTS

Turbocharged Engine Rebuild & Swap, 2006 Subaru Legacy GT

2025 - present

- Diagnosed an internal engine failure from running symptoms and a compression test, then chose to build a stronger engine rather than repair the original.
- Tore down the engine and had the block and heads machined, then rebuilt it myself with stronger internals (bearings, rings, head studs), upgraded cooling and oil systems, and an upgraded turbo and supporting bolt-ons to match the power goal.
- Validated the build through datalogging, monitoring and fault-checking it myself and working with a tuner to refine the calibration from logged data; now daily driven, with motorsport-style tune/calibration planned.

Tyre Temperature Sensing System, UC Motorsport (FSAE)

2026 - present

- Junior team member designing a standalone tyre-temperature monitoring system on a custom PCB built around an ESP32-S3 module, intended to feed a variable friction coefficient into the car's new traction control system. Took it from schematic through to a populated board with firmware written, now moving into testing.

Line Follower Robot, ENMT221 Mechatronics Design

2025

- Designed and 3D-printed the complete mechanical platform and drivetrain in SolidWorks for a robot built from scratch by the team (custom PCB, sensors and motor control), and took part in the on-track testing and tuning.
- Set a practice lap quick enough to have won that year's event; finished 3rd, 0.14s off first, after a late motor failure.

RoboCup Autonomous Robot, ENMT301 Mechatronics System Design

2026 - present

- Team of three building a fully autonomous robot for a timed arena challenge: navigating and avoiding obstacles, collecting target weights, and returning them to base using homing logic and control.
- Leading mechanical design and working across electrical and embedded systems; proposed the design concept the team chose to build. Build in progress: ramp and drum intake through multiple iterations, ToF sensing linked to drive control for obstacle response, and a sorting mechanism now in development.

SKILLS

Mechanical & production: Diagnostics & repair, fabrication, MIG & arc welding, machine tool operation (mill, lathe), structural verification, CAD (SolidWorks incl. FEA, Fusion 360), CAM (3D printing, laser cutting, CNC), MATLAB & Simulink

Electronics & hardware: Soldering, auto & marine wiring, PCB design (KiCad), LTSpice, surface-mount pick and place (with reflow)

Controls & embedded: PLC (ladder logic), embedded software (C, Python), firmware development, control theory, performance tuning, datalogging

Business & communication: Small business management, sales, bilingual (English/Mandarin), consultation, contractor coordination, social media marketing, technical reporting